

2.electrical damping



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A **magnet** suspended by a spring will display simple **harmonic motion** when displaced. If the magnet oscillates within a coil connected to a resistor, its motion will be **damped**. Investigate the factors that affect the damping.



Introduction

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Theoretical Analysis

Experiment

Conclusion

Overview

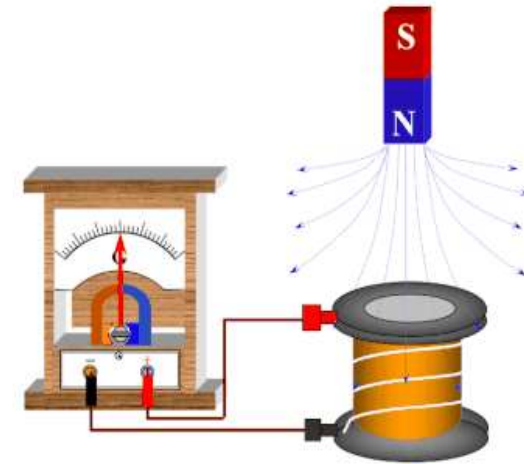
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Problem Analysis

A **magnet** suspended by a spring will display simple **harmonic motion** when displaced. If the magnet oscillates within a coil connected to a resistor, its motion will be **damped**. Investigate the **factors** that affect the **damping**.



Physical nature: **electromagnetic** damping. Conversion between magnet's **mechanical energy** and **electrical energy** via induction and then dissipated as **heat** through the resistor.

Parameters

Magnetic moment μ_{total}

Number of turns N

Resistance R

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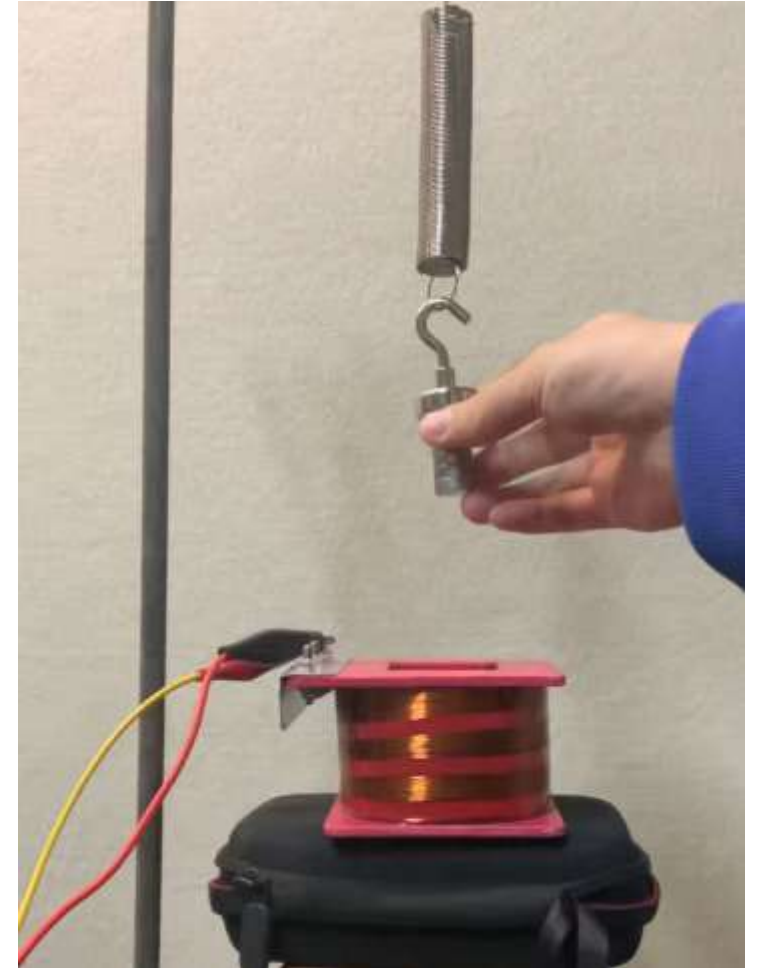
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Preliminary Experiment Phenomenon

Parameters

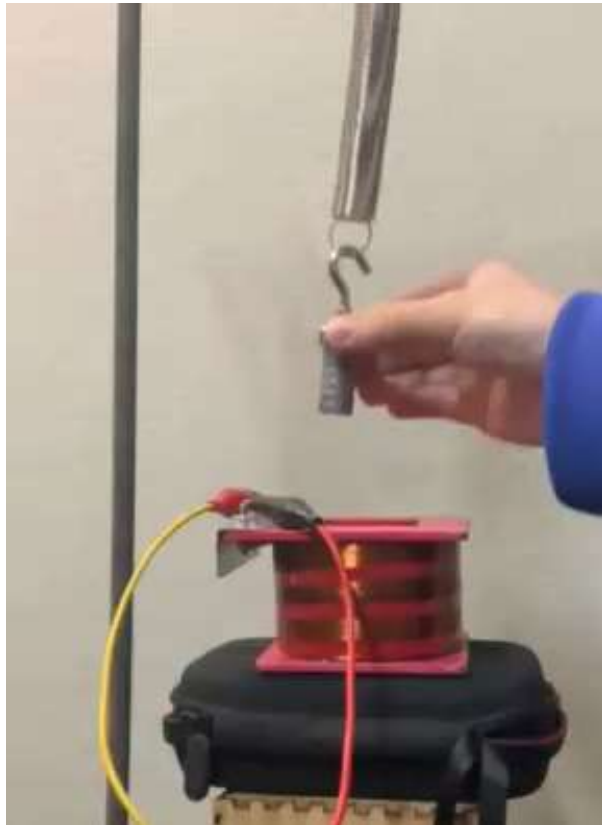
$R=10\ \Omega$, $N=400$ turns, $\mu_{total}= 27.02\ A \cdot m^2$,

- ! *Exponential Amplitude Decay as mechanical energy is continuously dissipated*
- ! *damping effect is slightly stronger when the magnet is near the coil than when it is away from it*
- ! *As the amplitude weakens, the rate of decreasing of amplitude also decreases*



Preliminaries

Number of turns



Increase the **number of coil turns**

- ① *Damping phenomenon increases*
- ② *Obvious damping appeared when the magnet approached the edge of the coil*

Preliminaries

Number of magnets



Increase the **number of magnets**

- ① *Vibration frequency decreases*
- ② *Damping phenomenon increases*

Introduction

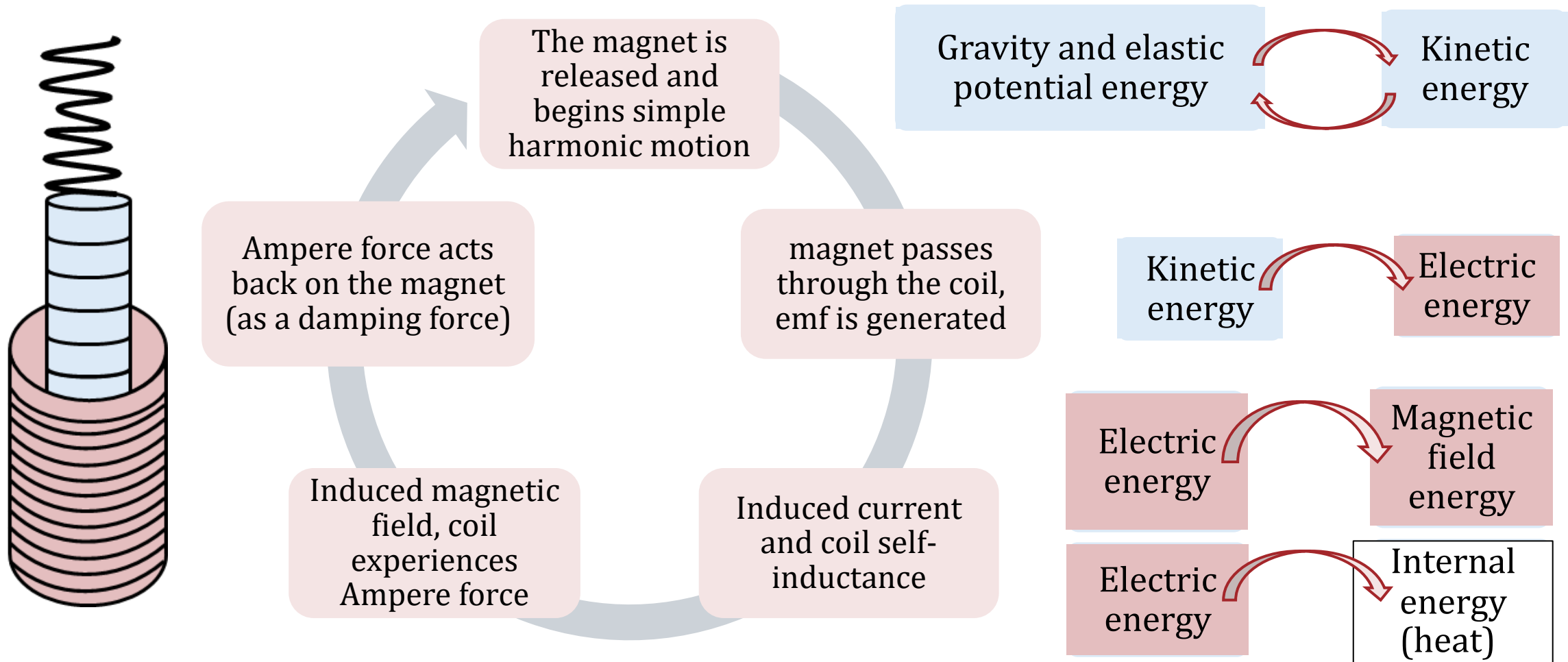
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Qualitative Analysis



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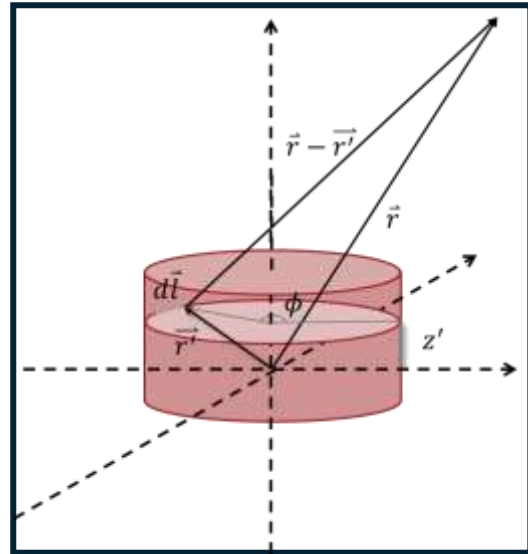
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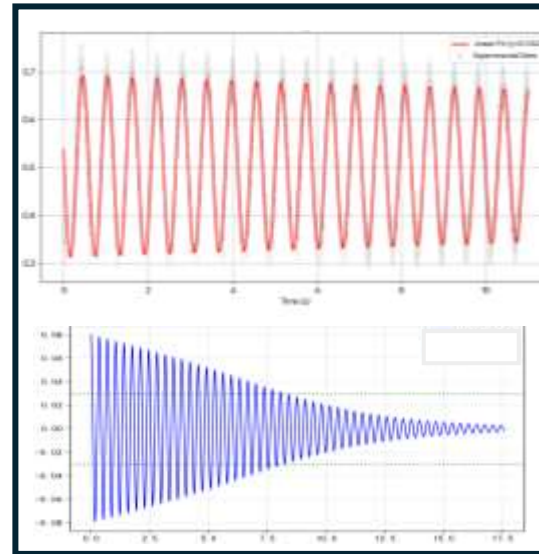
Overview

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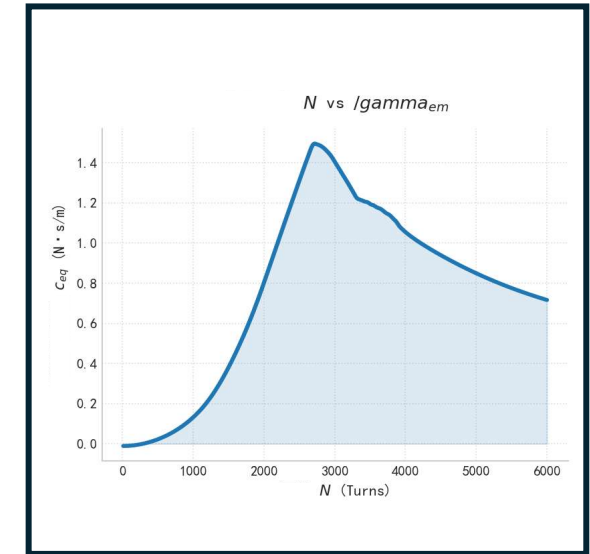
Theoretical analysis



Magnetic Field

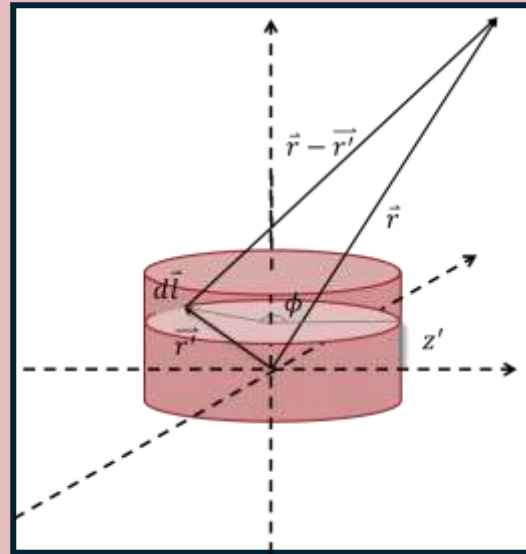


Kinematic Analysis

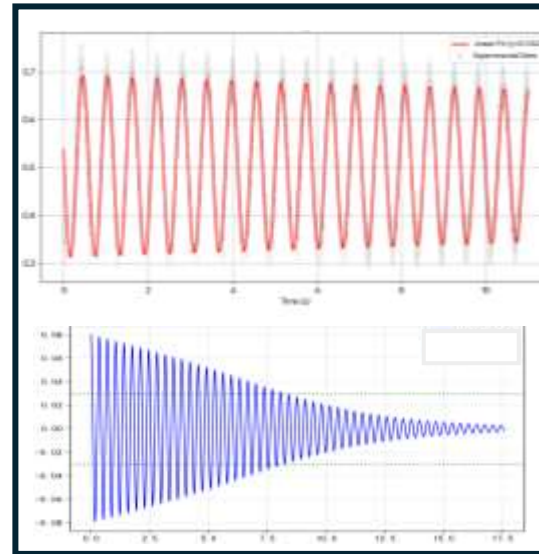


Parameter Analysis

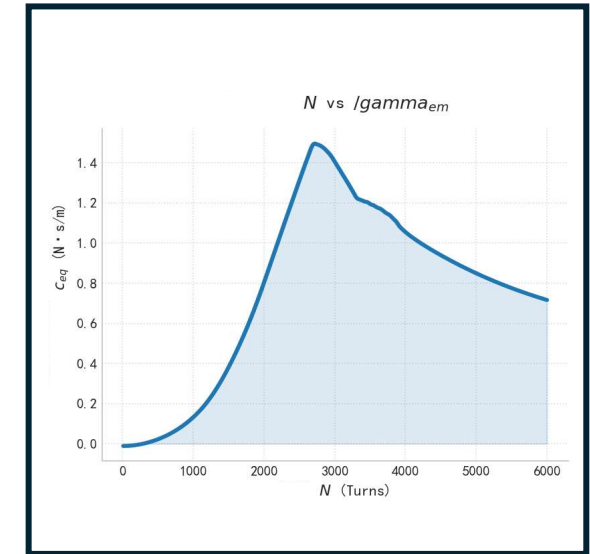
Theoretical analysis



Magnetic Field



Kinematic Analysis

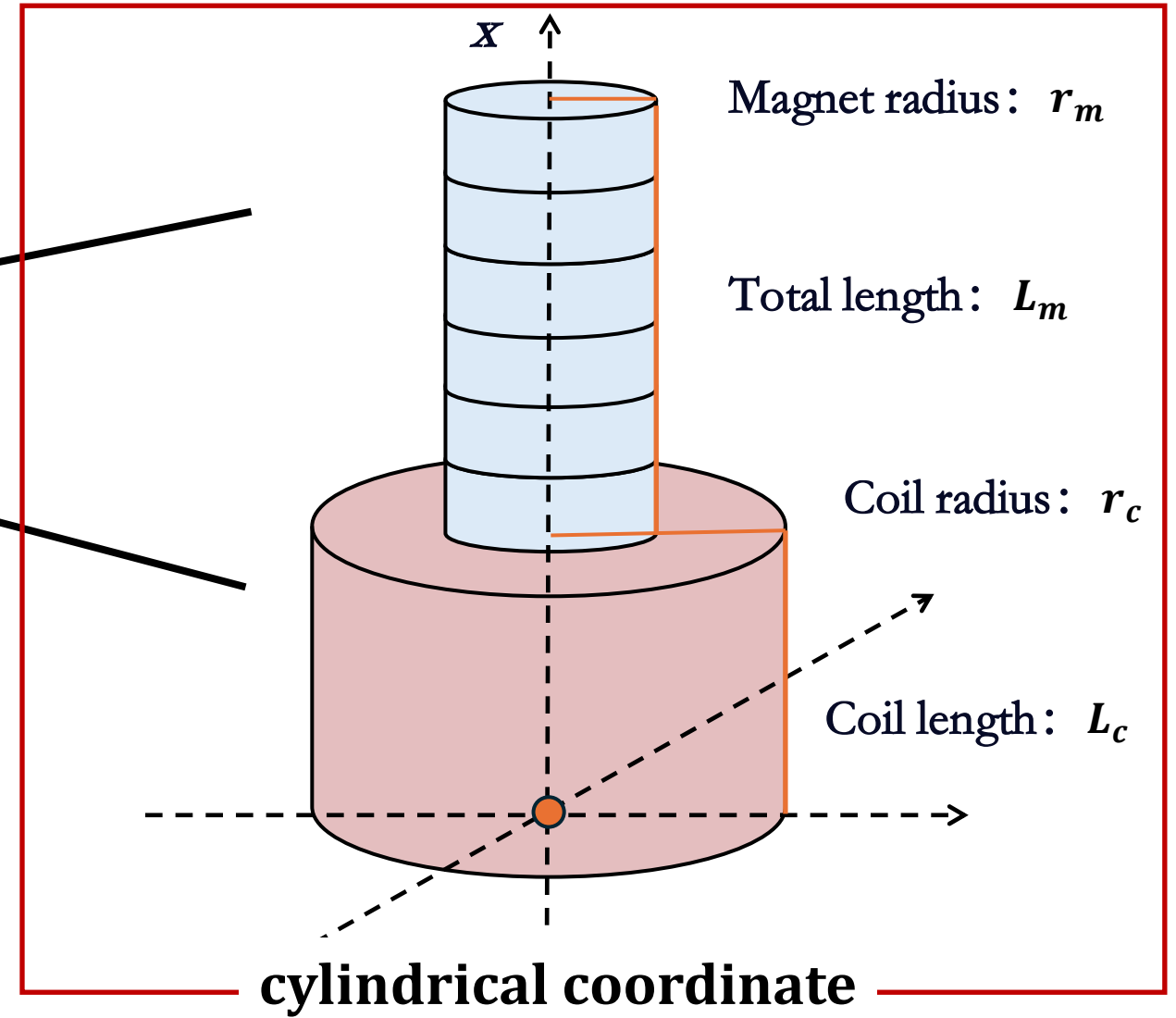
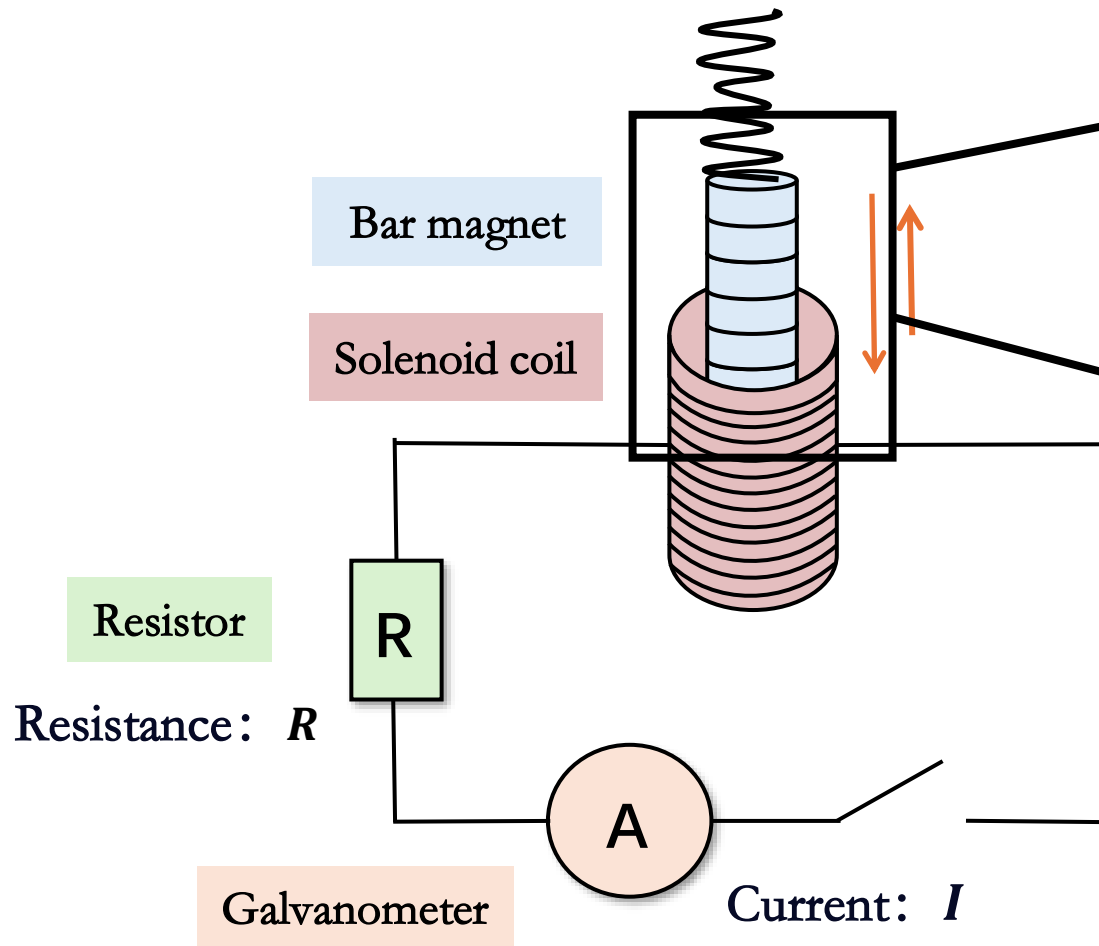


Parameter Analysis

Basic Assumptions

- 1 The spring is a linear spring
- 2 Motion occurs along a one-dimensional axis.

Reference Frame



center of the solenoid bottom as the origin

Introduction

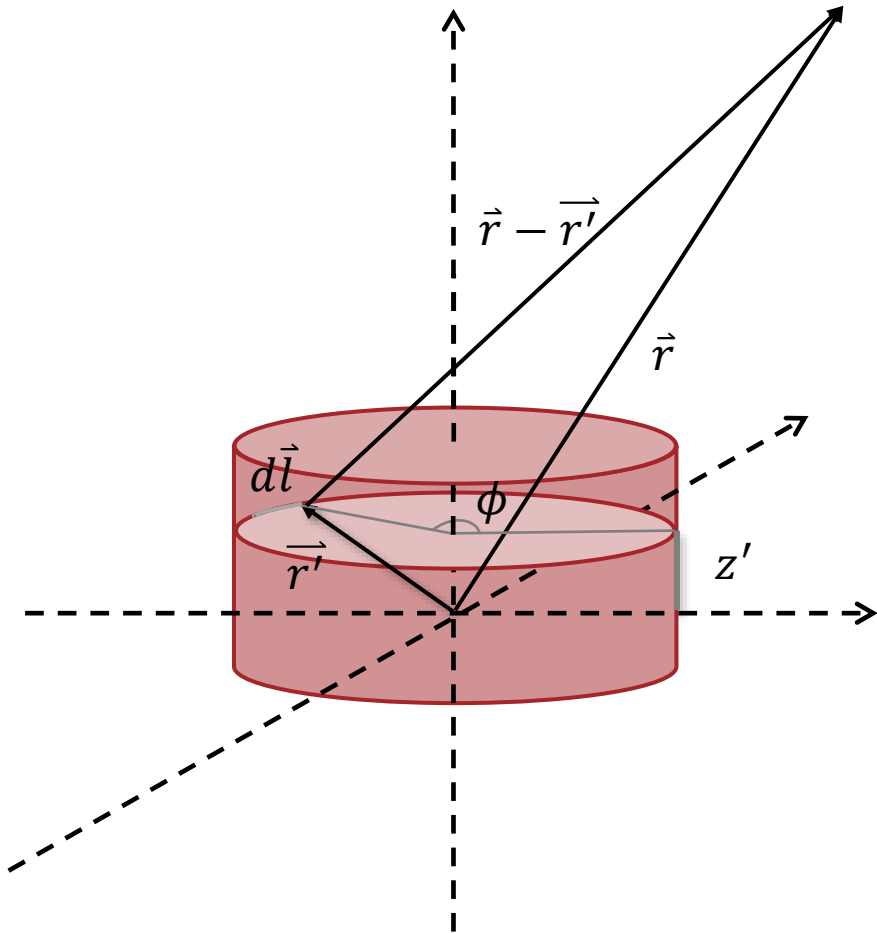
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Current Cylinder



Biot-Savart Law

$$\vec{r}' = (R \cos \phi, R \sin \phi, z')$$

$$d\vec{l}(\phi) = R d\phi (-\sin(\phi), \cos(\phi), 0)$$

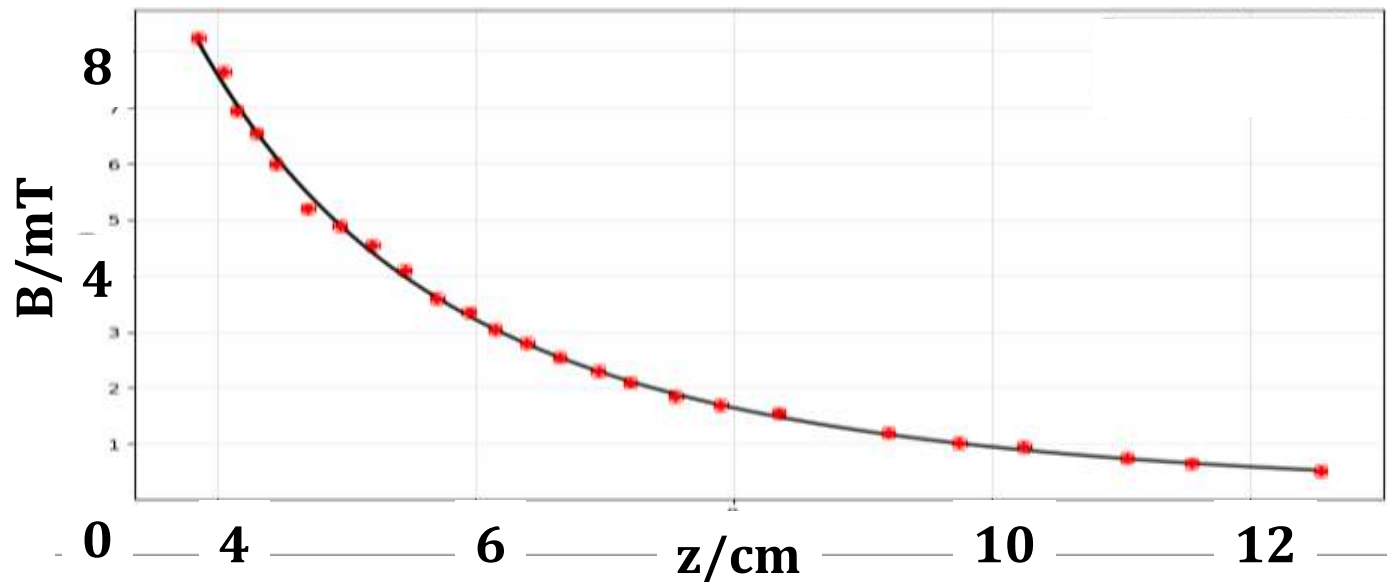
$$B(r) = \frac{\mu_0}{4\pi} \int \frac{I d\vec{l}' \times (\vec{r} - \vec{r}')}{|\vec{r} - \vec{r}'|^3}$$

$$\vec{B}(\vec{r}) = \frac{\mu_0 I}{4\pi} \int_{z'=0}^L \int_{\phi=0}^{2\pi} \frac{d\vec{l}(\phi) \times [\vec{r} - (R \cos(\phi), R \sin(\phi), z')]}{|\vec{r} - (R \cos(\phi), R \sin(\phi), z')|^3} dz'$$

Magnet Field

Axial Magnetic Field

$$B_z(z) = \frac{\mu_0 M}{2} \left[\frac{z + \frac{H}{2}}{\sqrt{\left(z + \frac{H}{2}\right)^2 + R^2}} - \frac{z - \frac{H}{2}}{\sqrt{\left(z - \frac{H}{2}\right)^2 + R^2}} \right]$$



Magnetization

$$\mathbf{M} = (1.12 \pm 0.04) \cdot 10^6 \text{ A/m}$$

Equivalent Current

$$dI = K_b dz = M_{mag} dz$$

$$I_{eq} = \int_0^L dI = M_{mag} L$$

Introduction

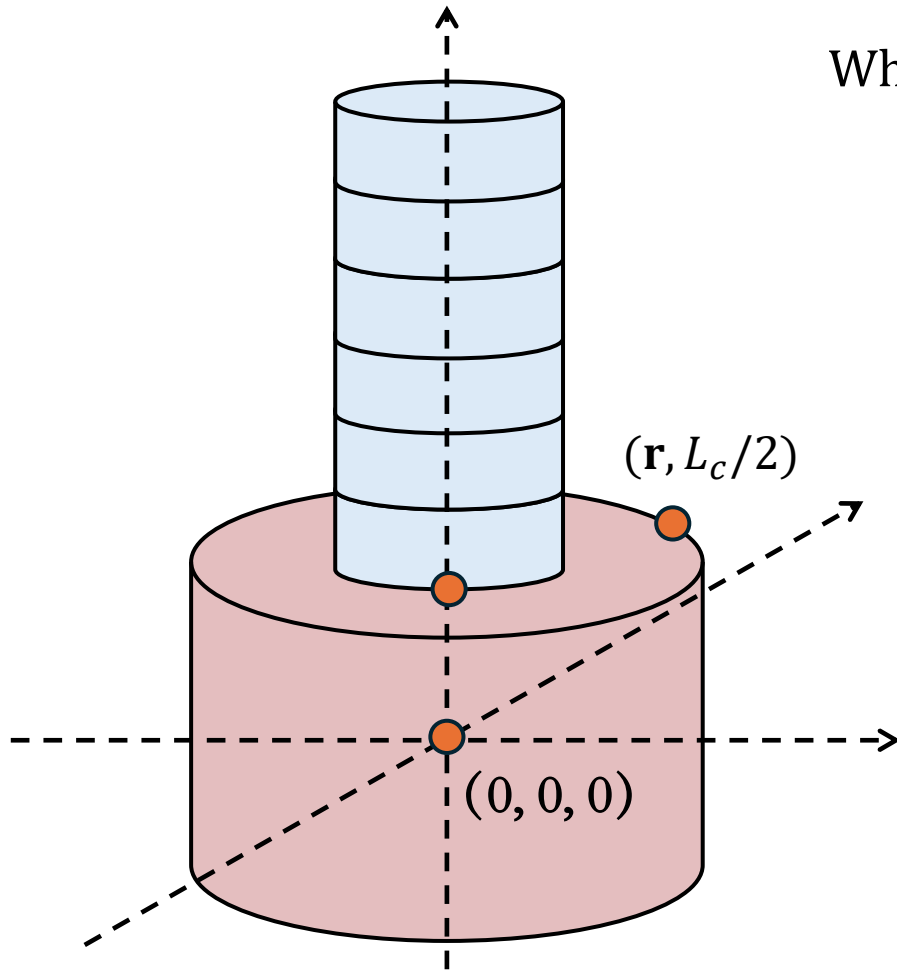
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Mutual Inductance Gradient



When the magnet bottom moves to position x :

total magnetic flux in the solenoid

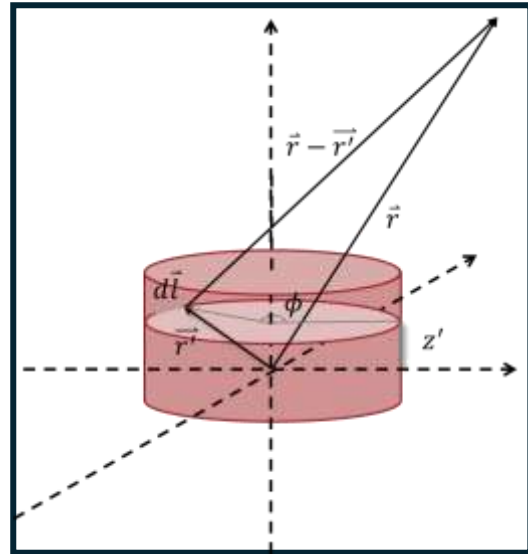
$$\psi((x, N, \mathbf{B})) = \int_0^{L_c} \left(\iint_{S(z)} \mathbf{B}(\mathbf{r}, z - x) \cdot d\mathbf{a} \right) ndz$$

mutual inductance gradient

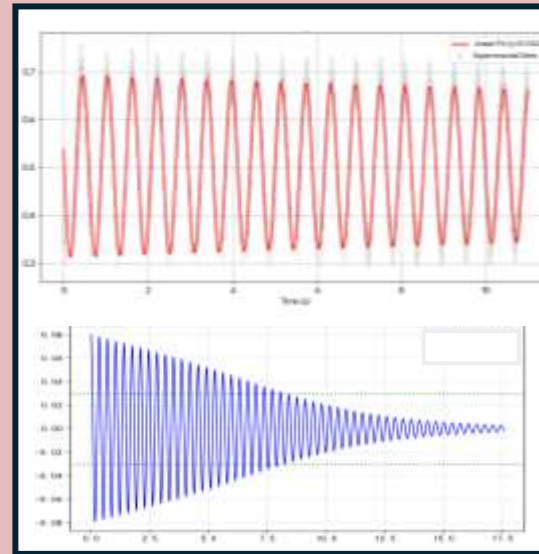
$$G((x, N, \mathbf{B})) = \frac{d\psi}{dx}$$

$$\varepsilon = \frac{d\psi}{dt} = \frac{d\psi}{dx} \cdot \frac{dx}{dt} = G((x, N, \mathbf{B})) \cdot \dot{x}$$

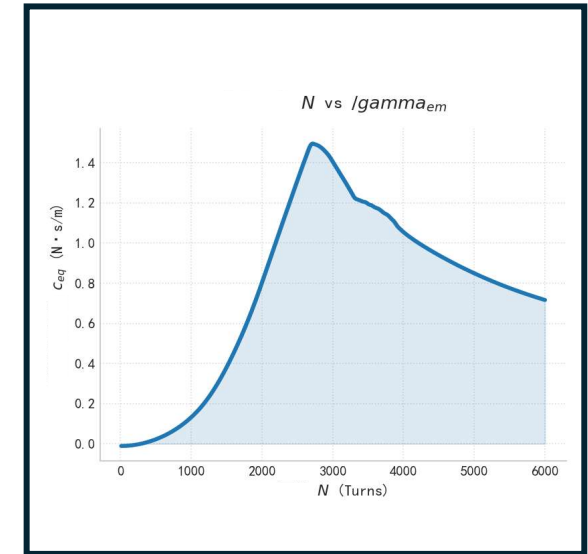
Theoretical analysis



Magnetic Field



**Kinematic
Analysis**



**Parameter
Analysis**

Total Energy

total energy of the system

$$E_{total} = \frac{1}{2}m\dot{x}^2 + \frac{1}{2}k(x - x_0)^2 + mgx + \frac{1}{2}LI^2$$

Mechanical
energy

Self-inductance
Magnetic Field
Energy

Without external work,

energy change

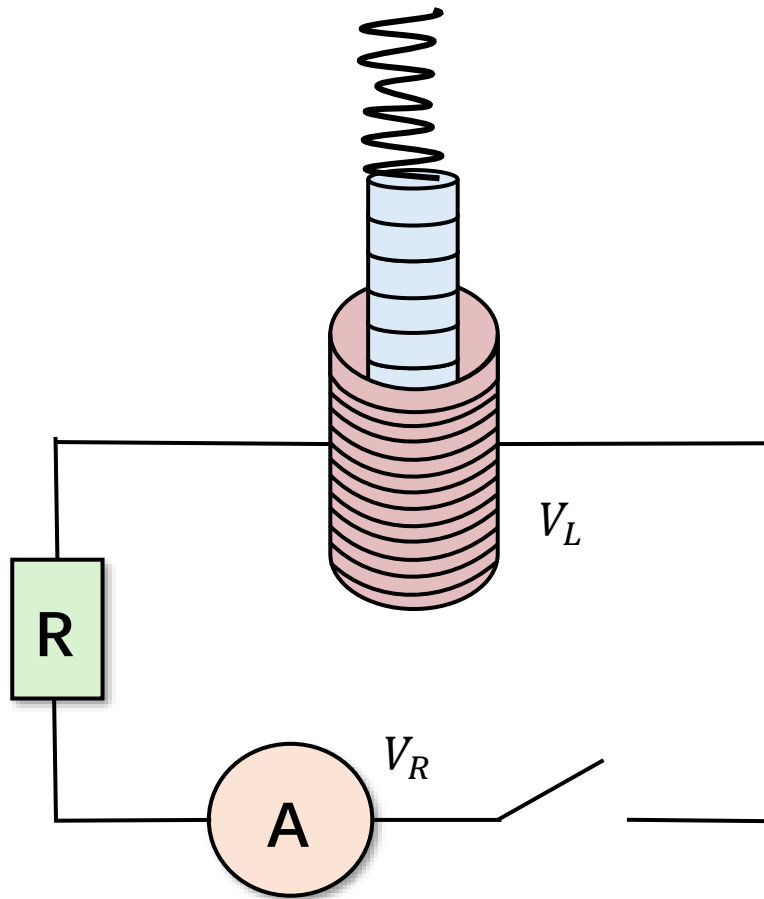
$$\frac{dE_{total}}{dt} = -P_{heat} = -I^2R$$

$$m\dot{x}\ddot{x} + k(x - x_0)\dot{x} + mg\dot{x} + LI\dot{I} = -I^2R$$

To discuss internal energy conversion, set external damping to 0

$$\dot{x}(m\ddot{x} + k(x - x_0) + mg) + I(L\dot{I} + IR) = 0$$

Electric Circuit



$$\dot{x}(m\ddot{x} + k(x - x_0) + mg) + \boxed{I(L\dot{I} + IR)} = 0$$

According to Kirchhoff's laws:

$$\varepsilon - V_L - V_R = 0$$

$$G(x, N, \mathbf{B}) \cdot \dot{x} - L\dot{I} - IR = 0$$

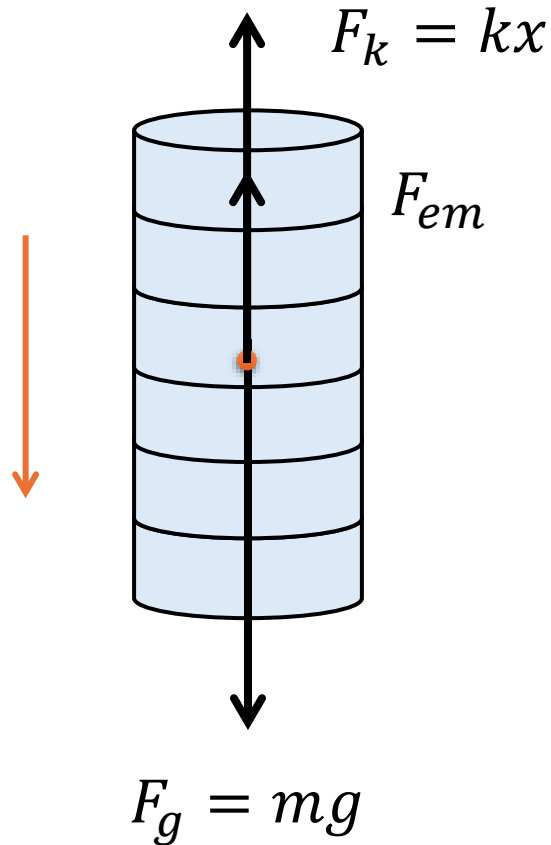
$$\boxed{G(x, N, \mathbf{B}) \cdot \dot{x} = L\dot{I} + IR}$$

R_r

$$R_{coil} = \rho \frac{l_{wire}}{S_{wire}} \approx \rho \frac{N \cdot (2\pi r_{coil})}{S_{wire}}$$

Mechanical Equilibrium

$$\dot{x}(m\ddot{x} + k(x - x_0) + mg) + I(L\dot{i} + IR) = 0$$



Taking upward as positive:

$$m\ddot{x} = -k(x - x_0) - mg + F_{em}$$

F_{em} is the Electromagnetic damping force

$$F_{em} = -(m\ddot{x} + k(x - x_0) + mg)$$

Damping Force

Combine the equations $\dot{x}(m\ddot{x} + k(x - x_0) + mg) + I(L\dot{i} + IR) = 0$

$$F_{em} = IG(x, N, \mathbf{B})$$

Scheme 1: **inductance is ignored**, analytical solution:

induced current $\frac{(G(x, N, \mathbf{B})) \dot{x}}{R}$

Damping expression

$$F_{em} = -\frac{(G(x, N, \mathbf{B}))^2}{R} \cdot \dot{x}$$

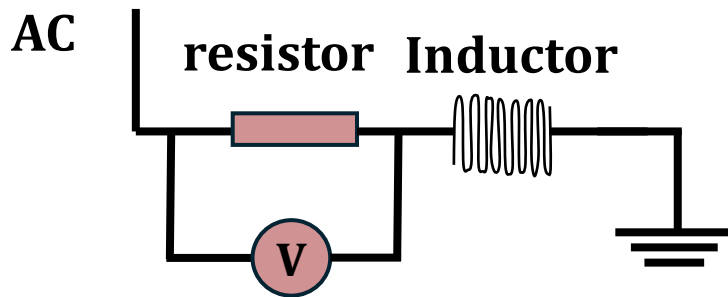
Electromagnetic damping coefficient

$$\gamma_{em} = \frac{(G(x, N, \mathbf{B}))^2}{R}$$

Inductance

Inductance measurement

RL Series Circuit Method



$$I = \frac{U_R}{R} \quad X_L = \frac{U_L}{I} \quad L = \frac{X_L}{2\pi f}$$



Inductance

$$L = 150 \pm 15 \text{ mH}$$

coupled differential equations

Scheme 2: **inductance is considered**, coupled differential equations

$$\left\{ \begin{array}{l} G((x, N, \mathbf{B})) = \frac{d\psi}{dx} \\ m\ddot{x} = -k(x - x_0) - mg + IG(x, N, \mathbf{B}) - 2m\gamma_{mech} \\ G(x, N, \mathbf{B})\dot{x} = L\dot{I} + IR \end{array} \right.$$

mutual inductance

Mechanical equation

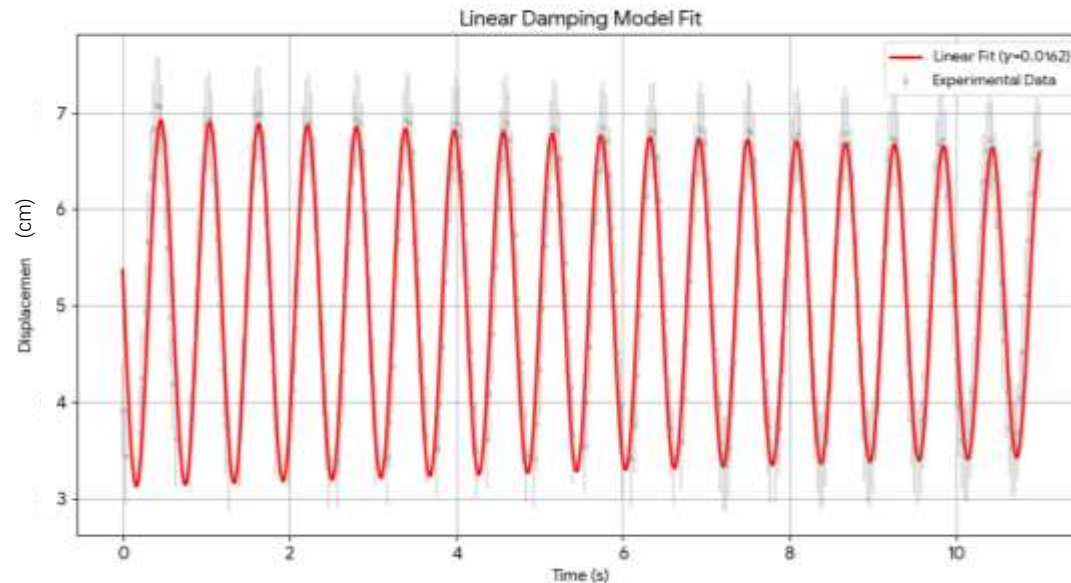
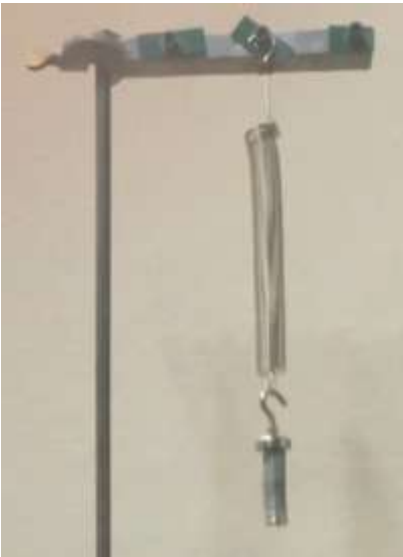
Kirchhoff's laws

Initial condition: $x(0) = 0.08\text{m}, v(0) = 0, I(0) = 0\text{A}$

Mechanical Damping Force

Mechanical damping measurement

Open circuit



Reynolds number

$$Re \approx 183$$

$\sqrt{\text{Linear damping}}$

**Mechanical
Damping Coefficient**

$$\gamma_{mech} = 0.0162 \pm 0.0005$$

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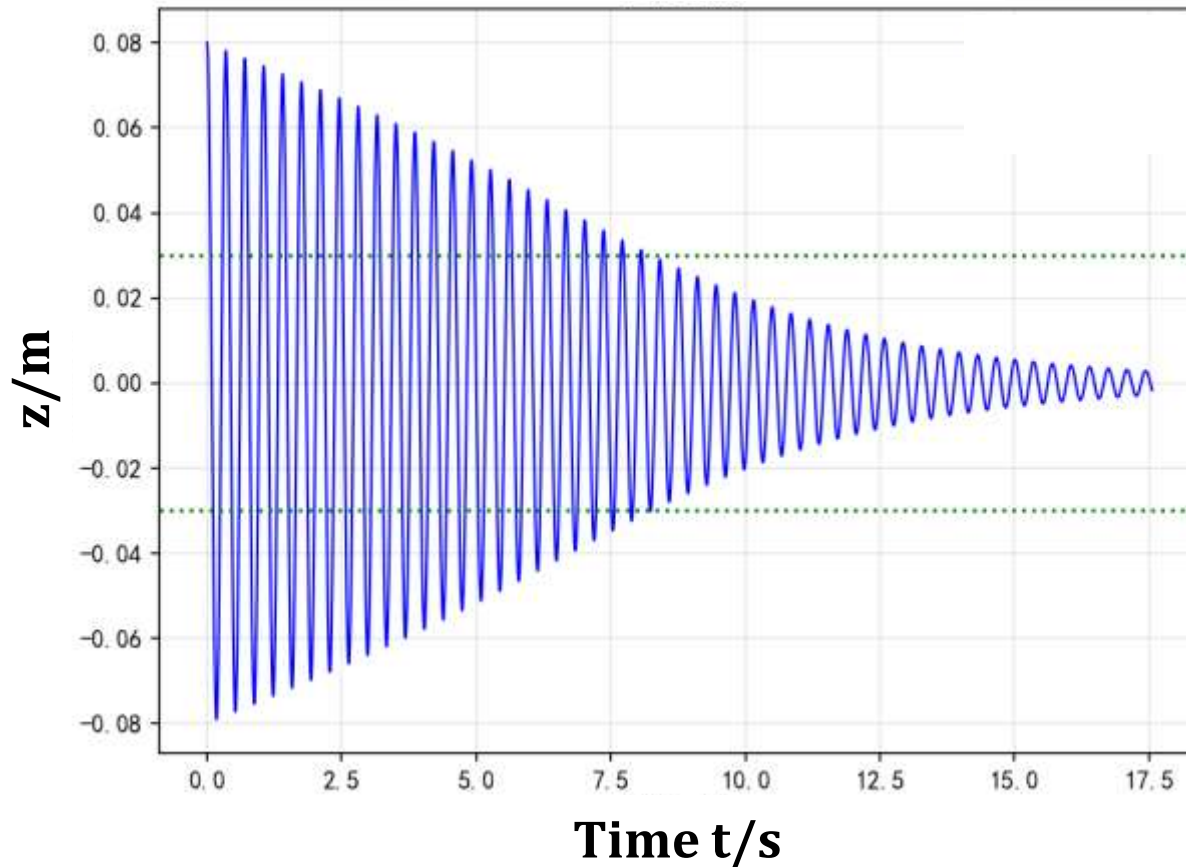
Conclusion

Motion Analysis

Parameters

$$R=3\ \Omega, N=400\ \text{turns}, \mu_{\text{total}}=27.02\ \text{A} \cdot \text{m}^2,$$

Displacement Analysis

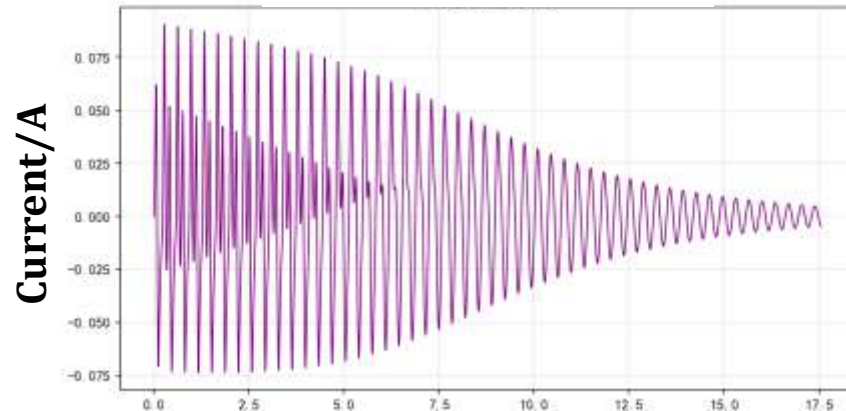


Solve using RK4

- ! Underdamped oscillation
- ! Decay rate of the envelope changes when magnets only reciprocates inside

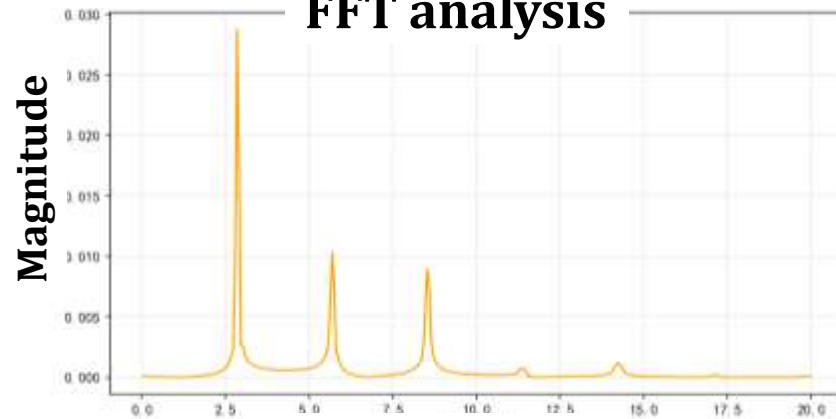
Current

Induced current



Time/s

FFT analysis



Frequency/Hz

Parameters

$$R=3\ \Omega, N=400\ \text{turns}, \mu_{\text{total}}=27.02\ \text{A}\cdot\text{m}^2,$$



Composed of multiple period motions

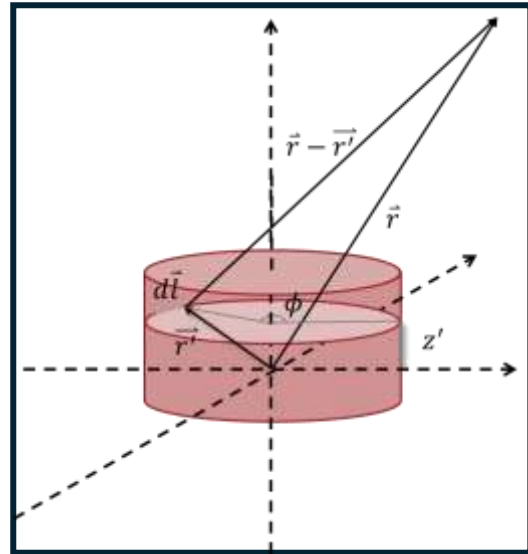


FFT analysis confirmed multi-frequencies superposition

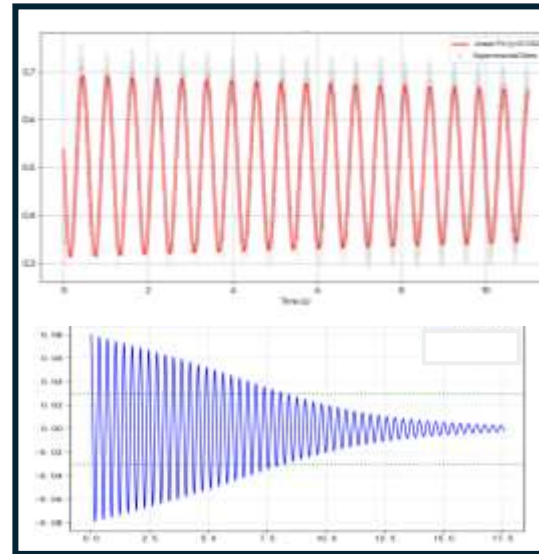


magnetic field possesses *high-order nonlinear* terms

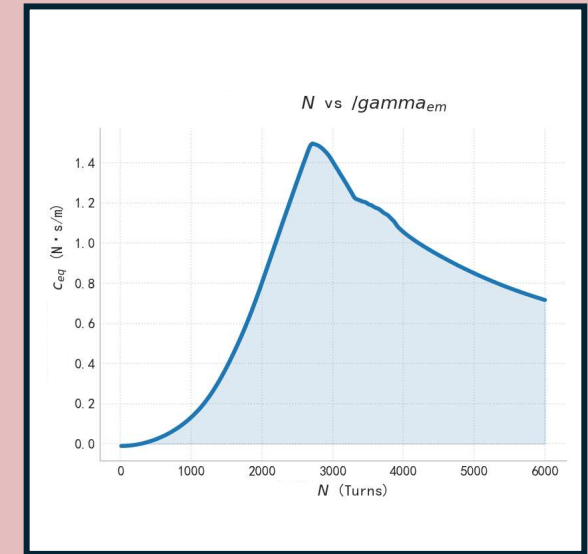
Theoretical analysis



Magnetic Field



Kinematic Analysis

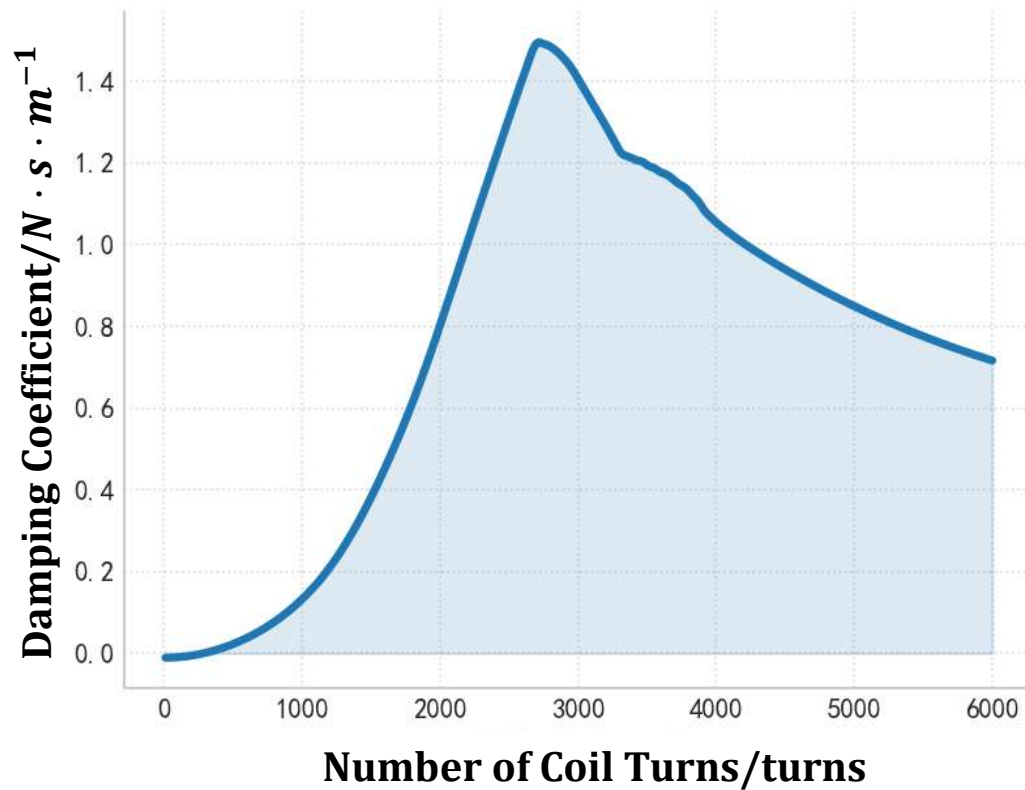


Parameter Analysis

Parameter Scanning

$$F_{em} = IG(x, N, B)$$

Influence of number of coil turns



①

Rapid rise phase (0–2700 turns):

Inductive reactance is small; ★ mutual inductance gradient, increasing with N

②

Peak point (2700 ± 10 turns): Energy dissipation maximum

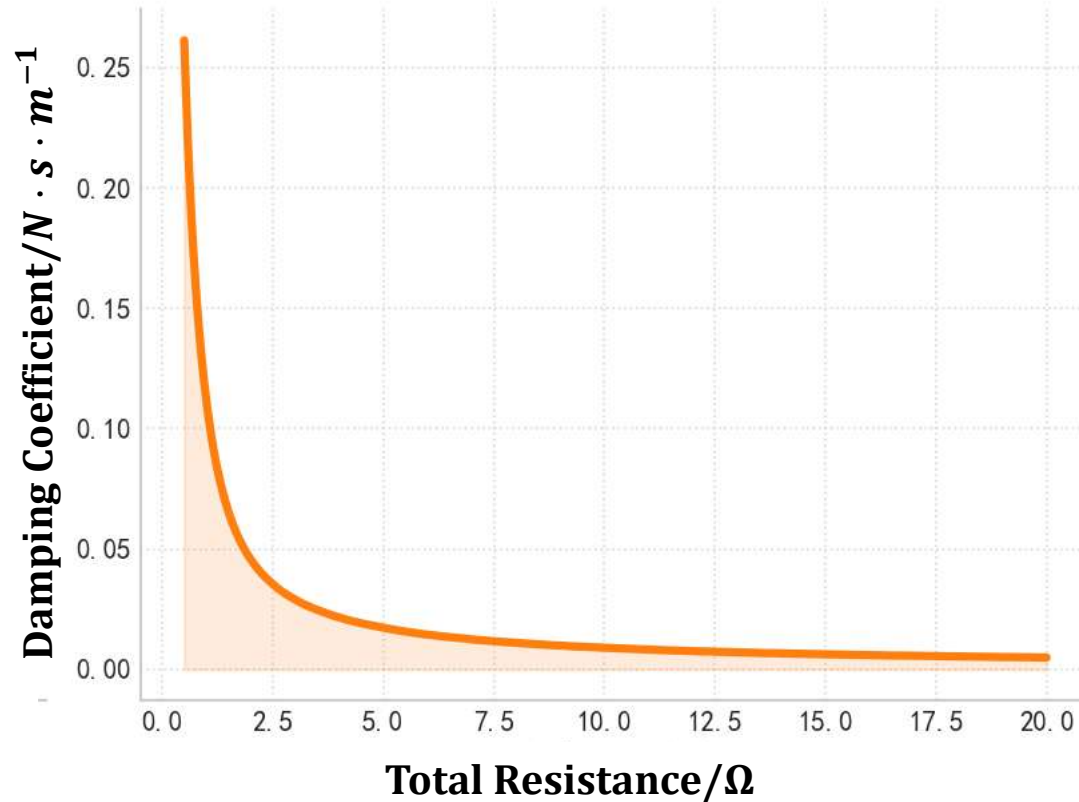
③

Slow decay phase (2700–6000 turns):

★ Inductive reactance increases rapidly; current, Ampere force, and damping all decrease.

Parameter Scanning $F_{em} = IG(x, N, B)$

Influence of Resistance



1

Approximate inverse relation

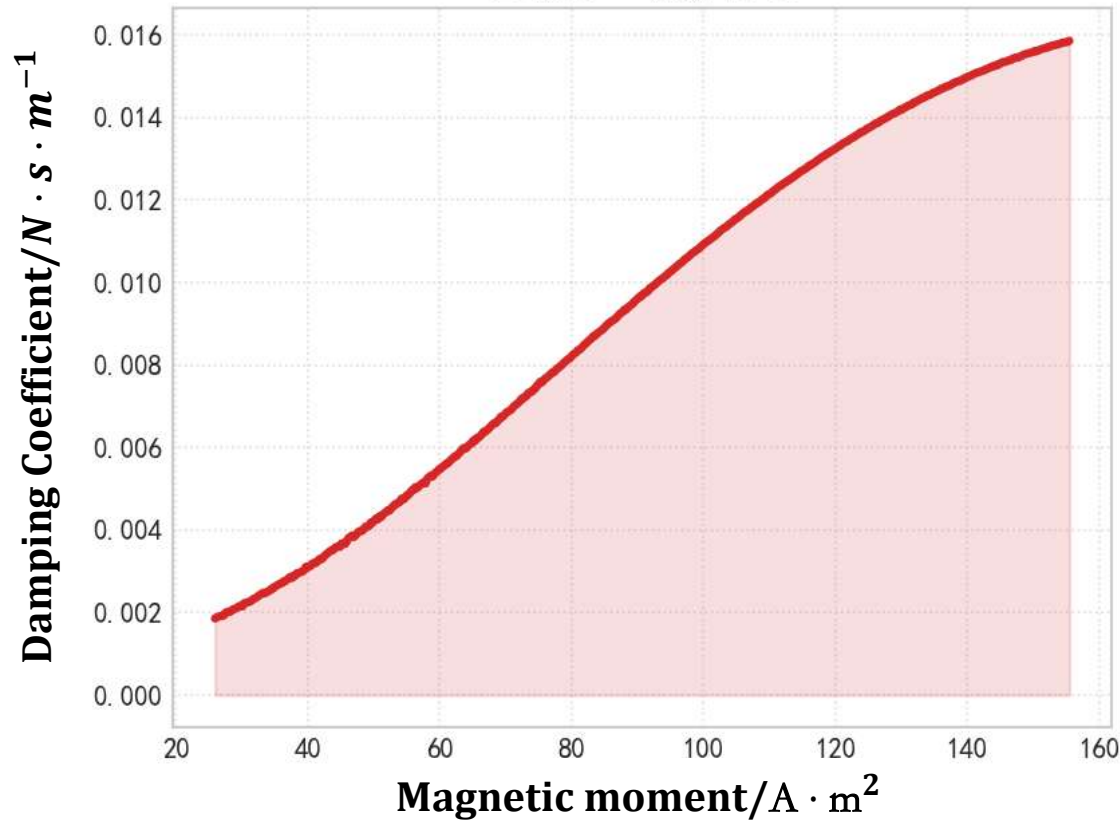
Current in the circuit is inversely proportional to total resistance

$$I \propto \frac{1}{R}$$

Parameter Scanning

$$F_{em} = IG(x, N, B)$$

Influence of magnetic moment



When the magnetic moment is small:
Quadratic growth

$$\varepsilon \propto m \Rightarrow I \propto m$$

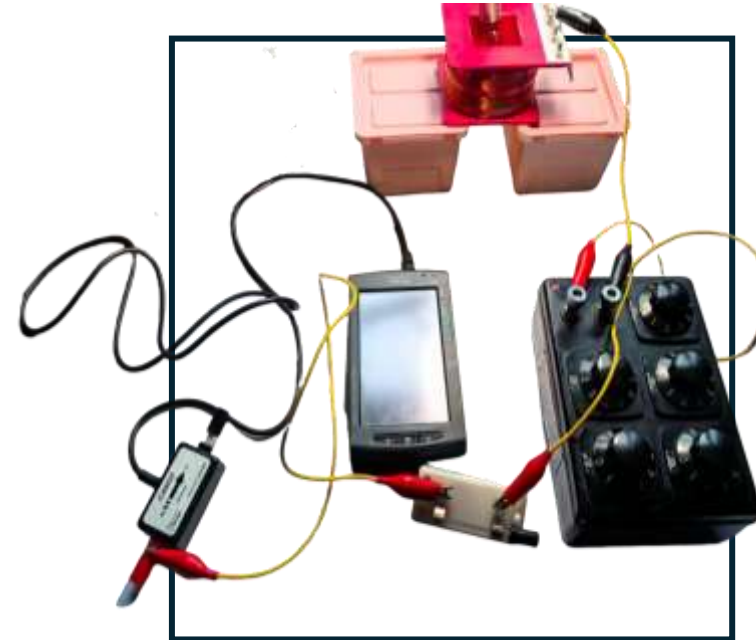
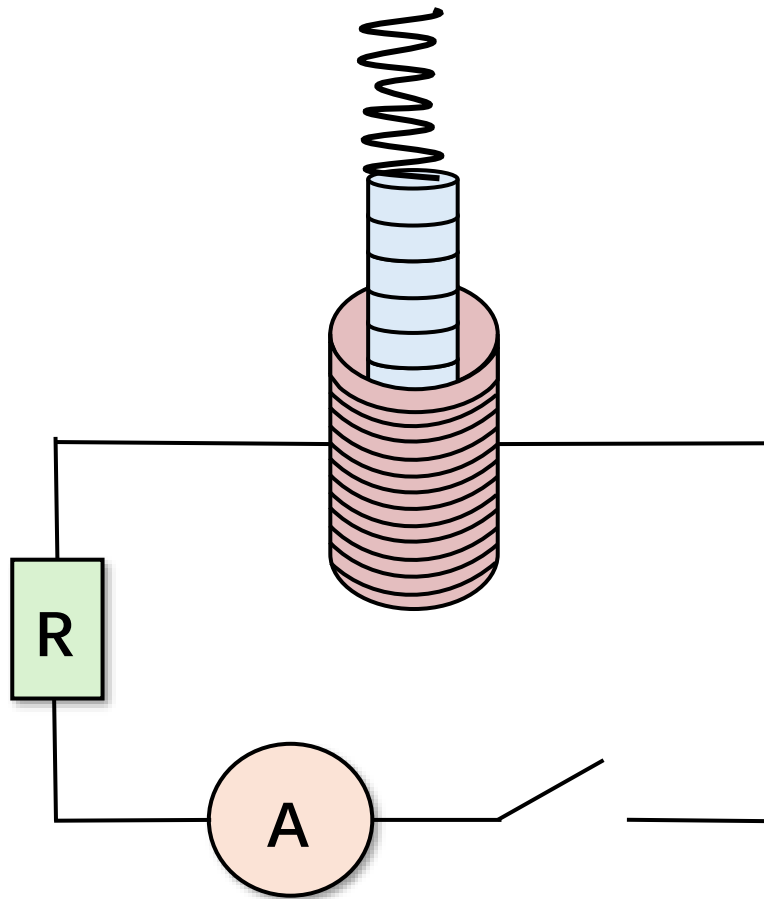
$$F_{em} \propto I \cdot m$$

When the magnetic moment is large:
mass $\uparrow$, growth rate of the damping
coefficient $\downarrow$

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Experimental Setup



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Materials and Apparatus

Coil turn density: 2666.67 turns/m
Cross-section area: 0.0045m^2

Coil

Switch

Displacement Sensor

spring

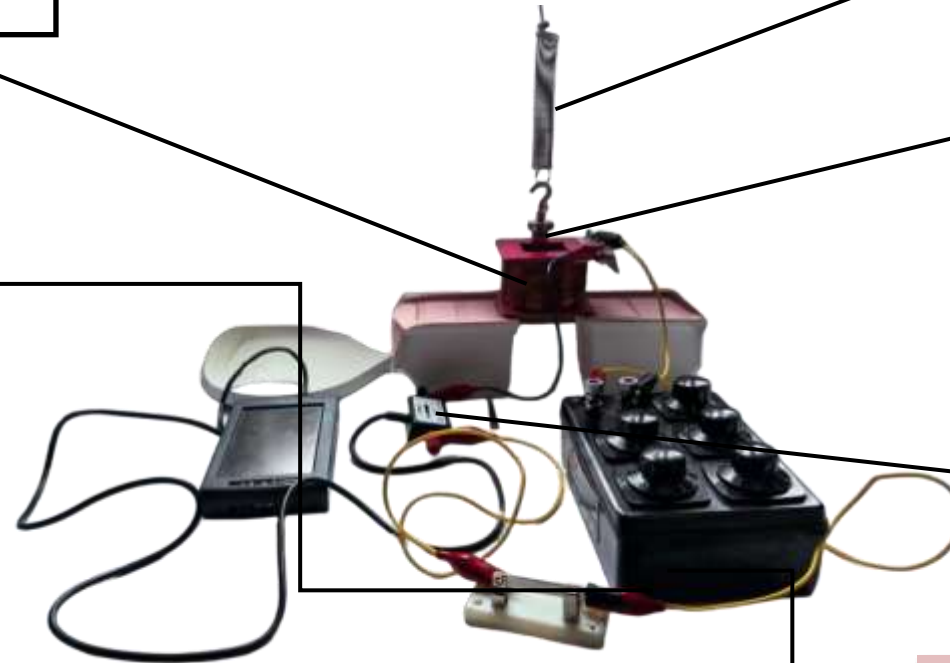
Stiffness: 16N/m
Total mass: 0.05kg

N52 cylindric magnet

radius: 0.008m
length: 0.012m

Galvanometer

Resistance Box



Introduction

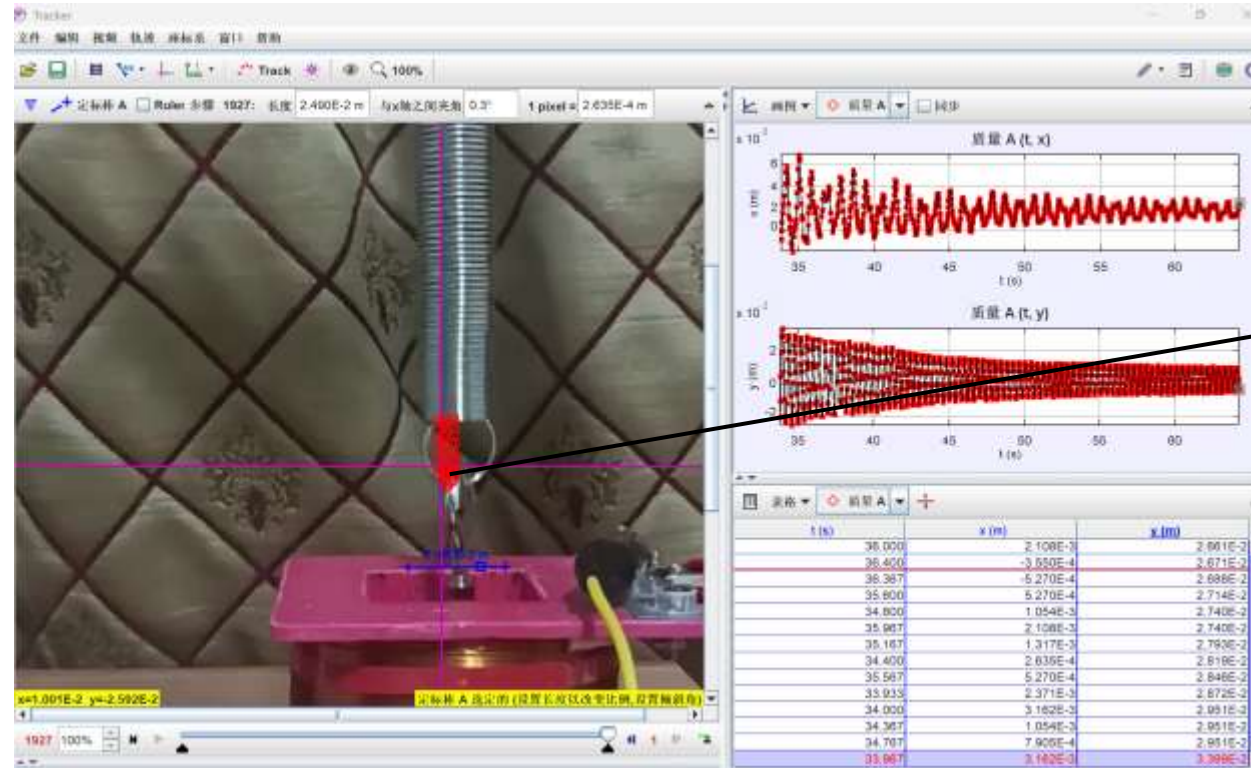
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Video Analysis/Tracker



top of the hook is used
as the equivalent
displacement of the
magnet

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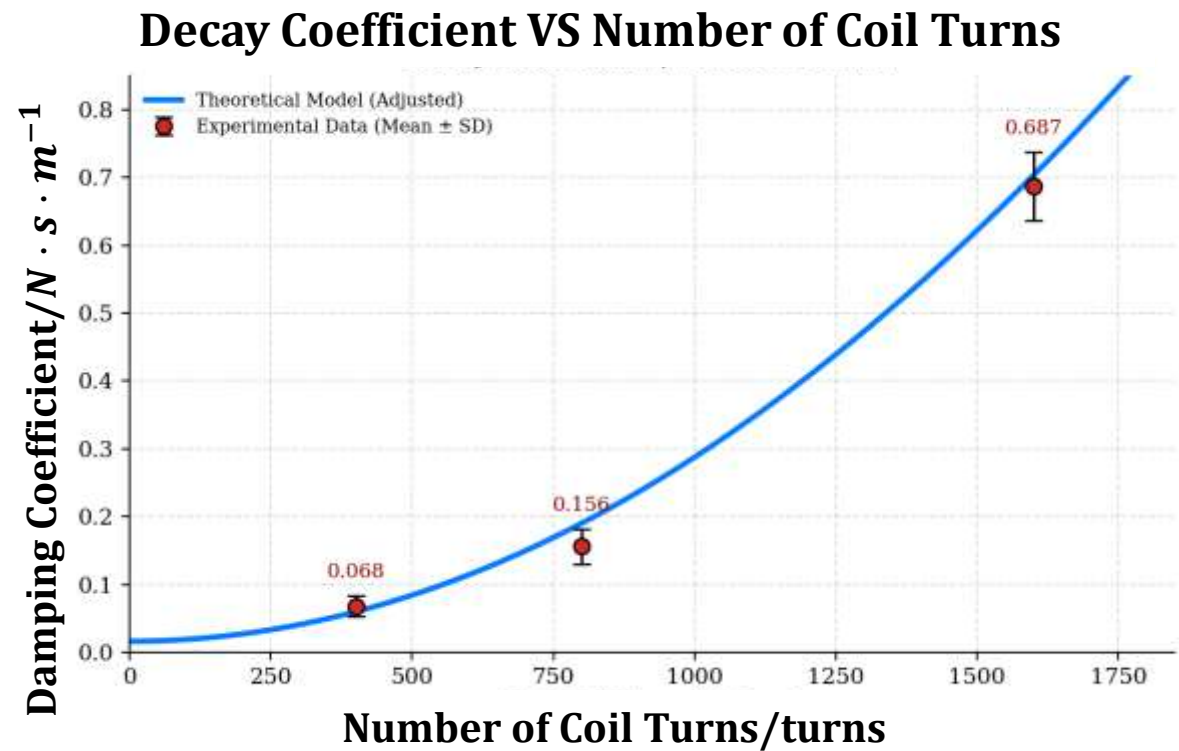
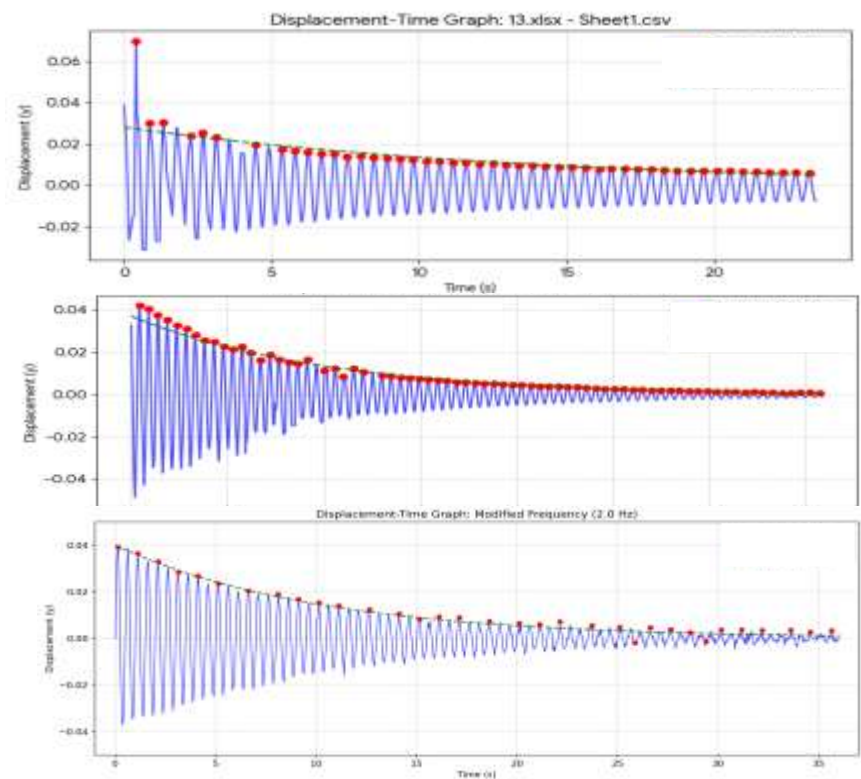
Theoretical Analysis

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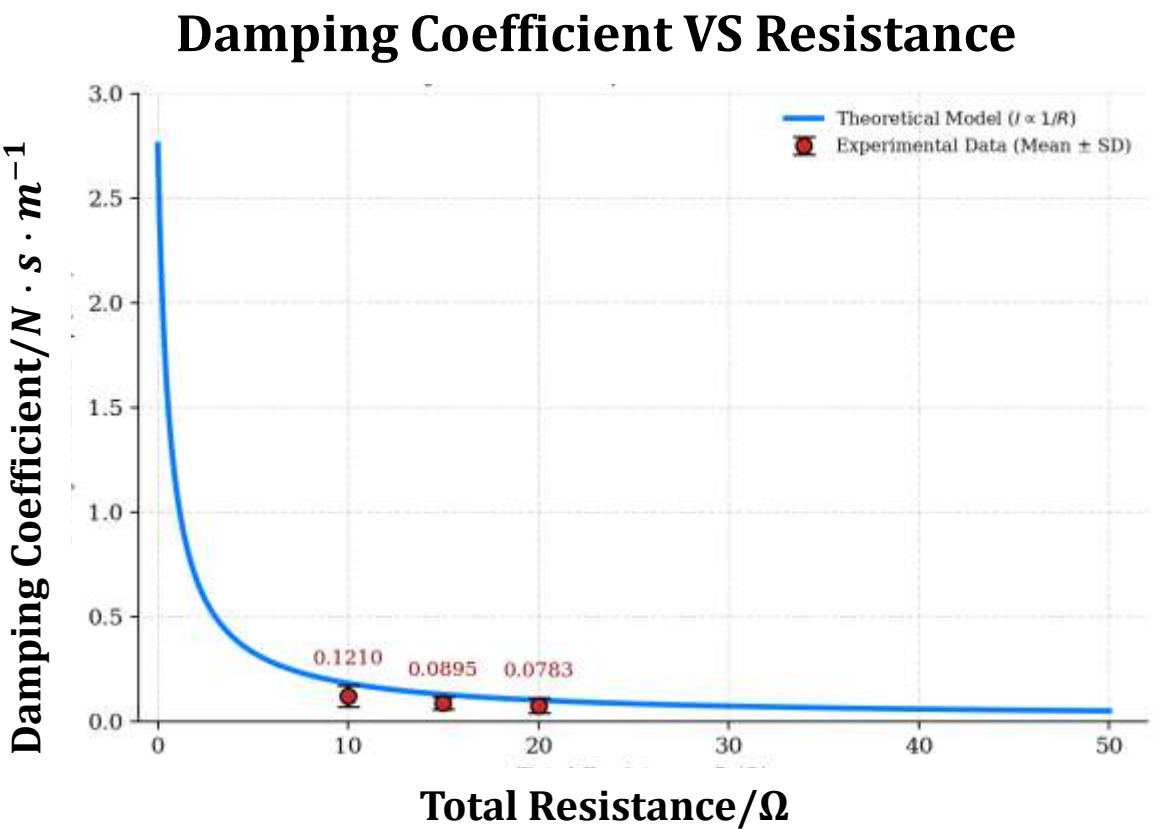
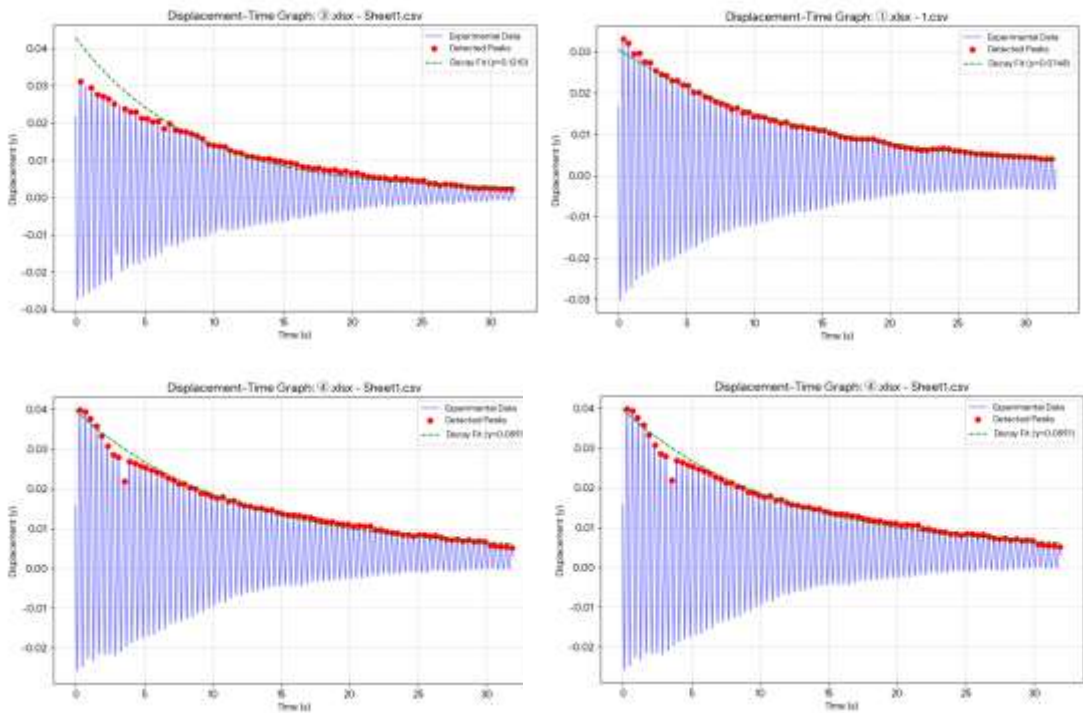
Experimental Results

Number of turns



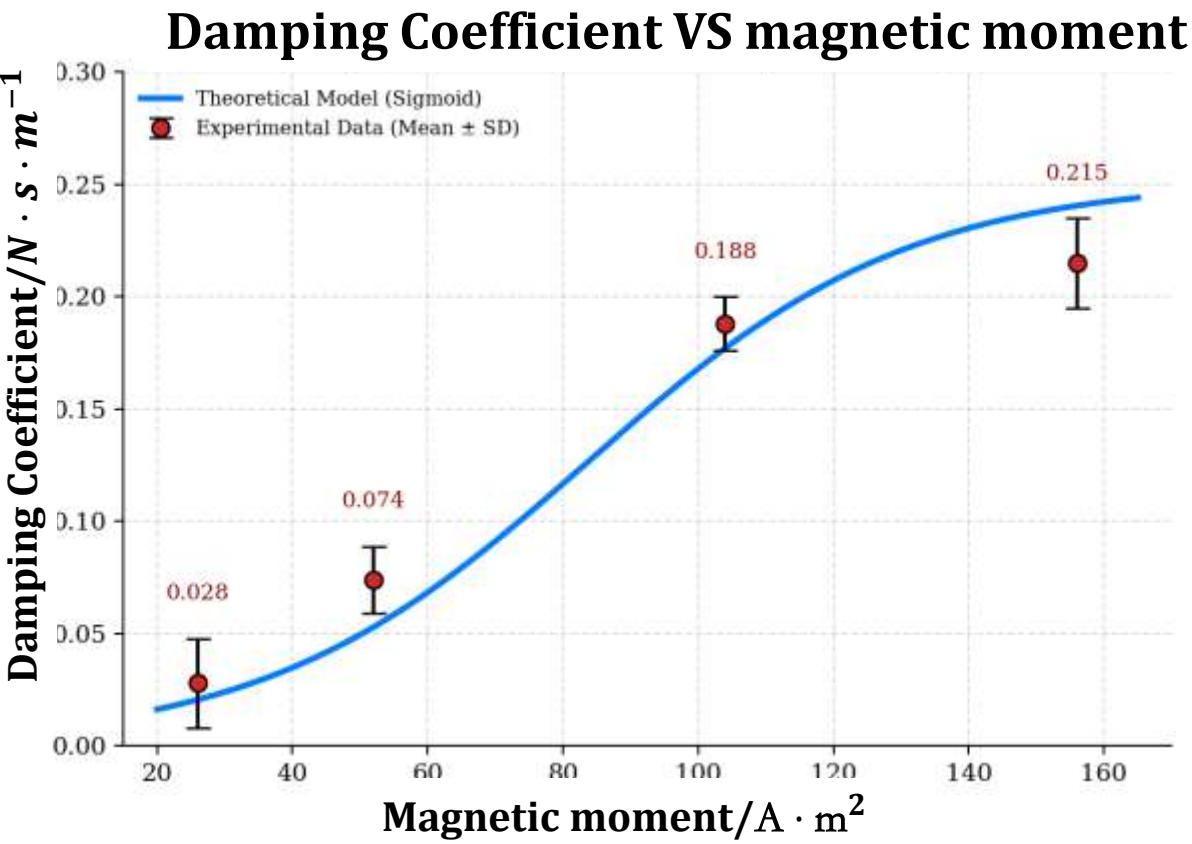
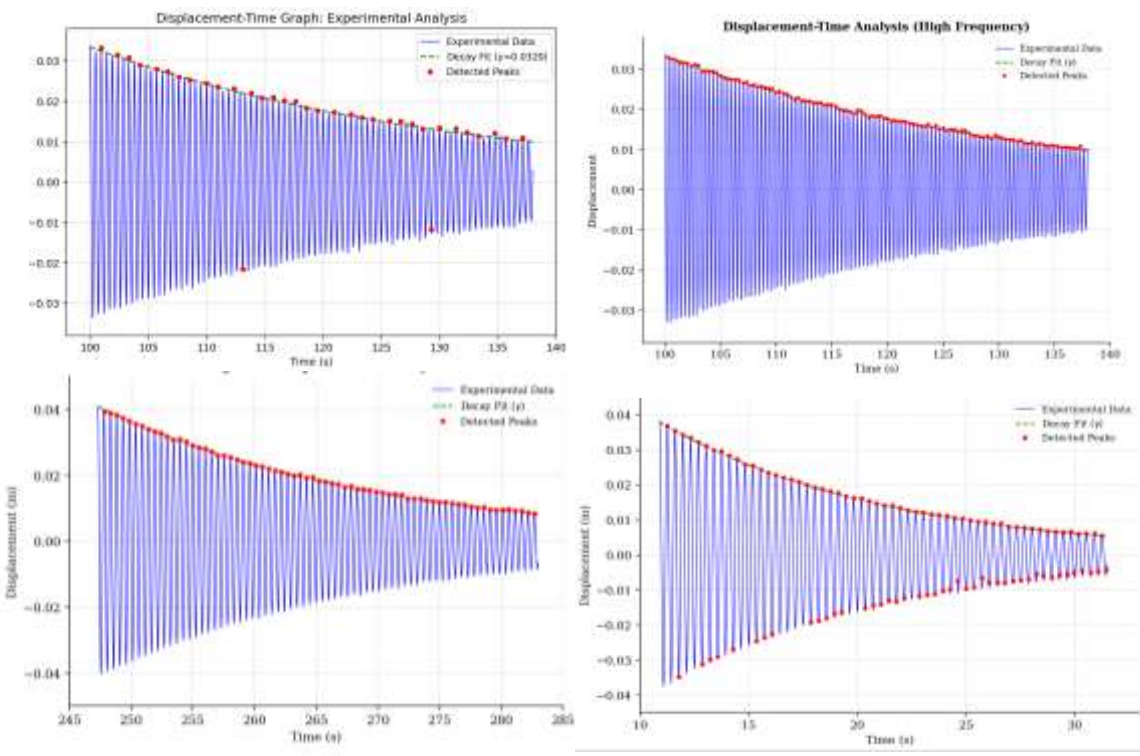
Experimental Results

Resistance



Experimental Results

Magnetic moment



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Conclusion

Physical nature

Mechanical energy converted into electrical and thermal energy via electromagnetic damping.

Well-aligned

Theoretical analysis

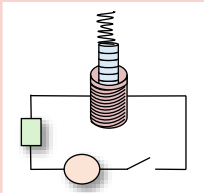
Experiment

Kinematic analysis

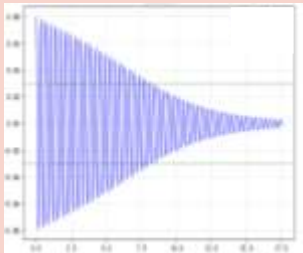
Parameter analysis

Multi-variable

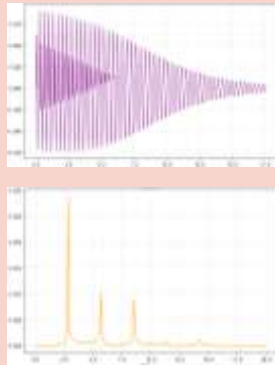
Quantitative model



Displacement



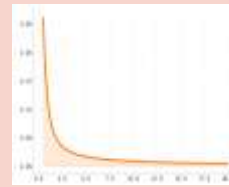
Induced Current



Coil turns



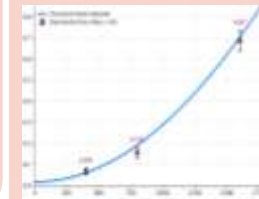
Resistance



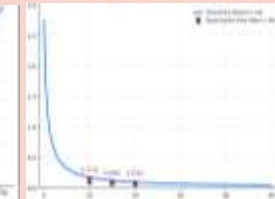
Magnetic moment



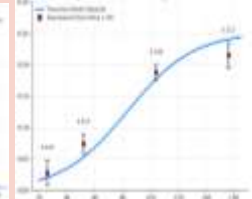
Coil turns



Resistance



Magnetic moment



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Error Analysis

Theoretical Model

1. Magnetization experimental measurement error 3.6%, $F_{em} \propto M^2$, damping coefficient error is 7.2%
2. Inductance measurement error 10%
3. Linear spring assumption: In large-amplitude oscillations, the spring may enter a non-linear region.
4. One-dimensional motion restriction: In the experiment, the magnet may have slight swinging or spinning, which changes the rate of change of magnetic flux $\frac{d\psi}{dt}$
5. Error in solving differential equations

Experimental Measurement

1. Mechanical damping error 3.3%
2. Video analysis error: A slight relative displacement may exist between the equivalent position and the actual position



Appendix

Introduction



Preliminaries



Theoretical Analysis



Experiment



Conclusion



Reynolds number

$$v_{max} = \omega A \approx 8.6 \times 0.02 = \mathbf{0.172 \text{ m/s}}$$

$$Re = \frac{\rho v D}{\eta} = \frac{1.2 \times 0.172 \times 0.016}{1.8 \times 10^{-5}} \approx \mathbf{183}$$

Experiment Design

- LabQuest
Display/Data
LoggerCurrent
Sensor Support
Stand
- Spring
- Identical High-
Strength
Magnets Coils
with Various
Turns
- Resistance Box
- Connecting
Wires Switch

Variables

Independent Variable:

Number of Coil Turns: 200, 800, 1600

Number of Magnets: 2, 4, 6, 8, 16

Circuit Resistance: 2Ω , 4Ω , 6Ω , 8Ω , 10Ω

Dependent Variables:

Magnet displacement and current

Controlled Variables

Released from the spring's natural length



Introduction

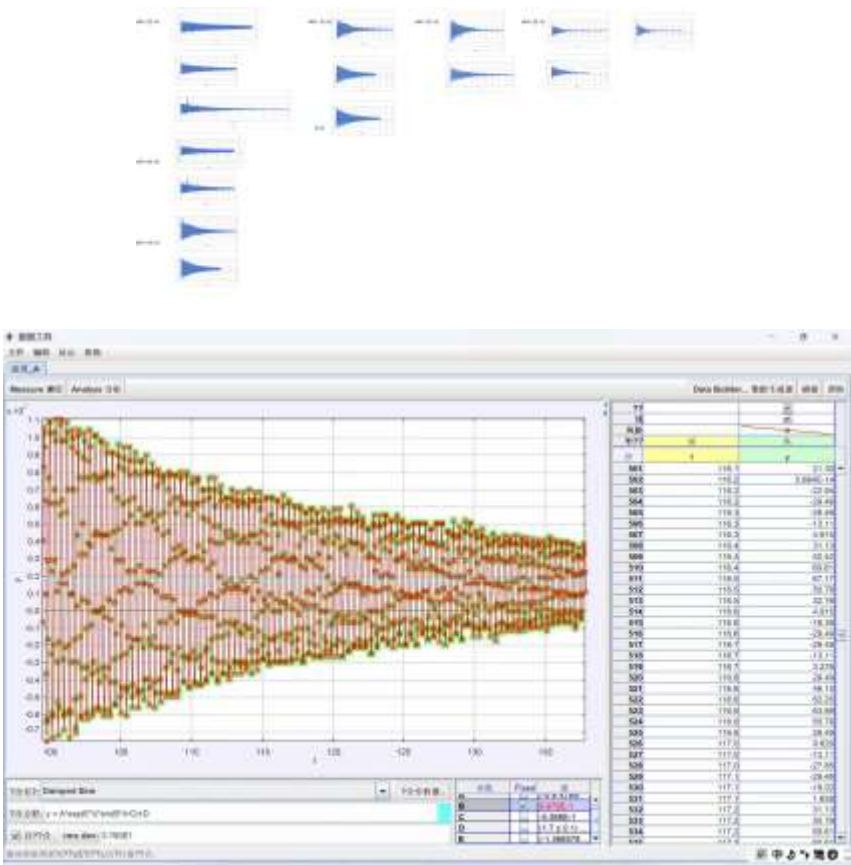
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Data Collection



名称	修改日期	类型
2磁铁, 2匝, 2欧完整-视频②.cmbf	2026-02-22 12:52	CMBL 文件
2磁铁, 2匝, 4欧完整-视频③.cmbf	2026-02-22 12:56	CMBL 文件
2磁铁, 2匝, 6欧完整-视频④.cmbf	2026-02-22 12:58	CMBL 文件
2磁铁, 2匝, 6欧完整-视频⑤.cmbf	2026-02-22 12:59	CMBL 文件
2磁铁, 2匝后半程-视频1.cmbf	2026-02-22 3:12	CMBL 文件
2磁铁, 2匝前程-视频2.cmbf	2026-02-22 3:14	CMBL 文件
2磁铁, 2匝前程-视频3.cmbf	2026-02-22 3:17	CMBL 文件
2磁铁, 8匝前程-视频4.cmbf	2026-02-22 3:19	CMBL 文件
2磁铁, 8匝前程-视频5.cmbf	2026-02-22 3:21	CMBL 文件
2磁铁, 16匝前程-视频6.cmbf	2026-02-22 3:24	CMBL 文件
2磁铁, 16匝前程-视频7.cmbf	2026-02-22 3:26	CMBL 文件
2磁铁; 匝数: 2.cmbf	2026-02-19 0:29	CMBL 文件
3.mp4	2026-02-22 14:03	MP4 文件
4.mp4	2026-02-22 14:03	MP4 文件
4磁铁, 2匝碰撞前-视频8.cmbf	2026-02-22 3:45	CMBL 文件
4磁铁, 2匝前程-视频9.cmbf	2026-02-22 3:47	CMBL 文件
4磁铁, 2匝前程-视频10.cmbf	2026-02-22 3:48	CMBL 文件
4磁铁, 8匝碰撞前-视频11.cmbf	2026-02-22 11:13	CMBL 文件
4磁铁, 8匝碰撞前-视频13.cmbf	2026-02-22 11:21	CMBL 文件
4磁铁, 8匝碰撞前-视频14.cmbf	2026-02-22 11:25	CMBL 文件
4磁铁, 8匝前程-视频12.cmbf	2026-02-22 11:15	CMBL 文件
6磁铁, 2匝, 有碰撞-视频15.cmbf	2026-02-22 11:48	CMBL 文件
6磁铁, 2匝完整-视频16.cmbf	2026-02-22 11:54	CMBL 文件
8磁铁, 2匝完整-视频17.cmbf	2026-02-22 12:26	CMBL 文件
8磁铁, 2匝完整-视频18.cmbf	2026-02-22 12:27	CMBL 文件
10磁铁, 2匝完整-视频19.cmbf	2026-02-22 12:31	CMBL 文件
10磁铁, 2匝完整-无视频19.cmbf	2026-02-22 12:45	CMBL 文件

Uncertainty Propagation

$$\frac{\Delta\mu}{\mu} = \frac{\Delta M}{M}$$

$$\Delta\mu = \mu \times \left(\frac{\Delta M}{M} \right)$$

$$\Delta\mu = 2.702 \times 0.0357 \approx 0.0965, \text{A} \cdot \text{m}^2$$